

Experiment No. 9

Experiment Name

Object Detection and Recognition using Deep Learning Models on Standard Image Datasets

Aim

To implement a deep learning-based object detection and recognition system using pre-trained models and evaluate its performance on standard image datasets for identifying and localizing multiple objects.

Purpose

Object detection is one of the most significant tasks in computer vision, combining object localization and object classification within an image. Unlike image classification, which predicts only the image category, object detection identifies the location of one or more objects by generating bounding boxes along with class labels.

In this experiment, students will utilize pre-trained deep learning models such as YOLOv8, SSD (Single Shot Detector), or Faster R-CNN to detect and recognize objects from standard datasets such as MS COCO, Pascal VOC, or Open Images Dataset. The experiment provides practical exposure to modern object detection frameworks widely used in autonomous vehicles, surveillance systems, healthcare, and smart city applications.

Prerequisites

- Basic knowledge of Digital Image Processing
- Familiarity with Python programming
- Python libraries: OpenCV (cv2), Ultralytics YOLO, PyTorch/TensorFlow, NumPy, and Matplotlib
- Understanding of Convolutional Neural Networks (CNNs)
- Jupyter Notebook, Google Colab, or Visual Studio Code

Steps

1. Install and import the required libraries, including OpenCV and the chosen deep learning framework (e.g., Ultralytics YOLO or TensorFlow).
2. Load a pre-trained object detection model such as YOLOv8, SSD, or Faster R-CNN.
3. Download or access images from a standard dataset such as MS COCO, Pascal VOC, or Open Images Dataset.

4. Perform any necessary image preprocessing, including resizing and normalization, before inference.
5. Apply the object detection model to identify and localize objects within the selected images.
6. Display the detected objects with bounding boxes, class labels, and confidence scores.
7. Evaluate the detection results by analyzing correctly detected objects, false detections, and missed objects.
8. Compare the performance of different object detection models in terms of detection accuracy and inference speed (if multiple models are used).
9. Analyze the suitability of the selected model for real-world applications such as autonomous driving, surveillance, retail analytics, or industrial inspection.
10. Summarize the observations and discuss the advantages and limitations of deep learning-based object detection techniques.

Questions

1. Differentiate between image classification, object detection, and image segmentation.
2. Explain the working principle of the YOLO (You Only Look Once) object detection algorithm.
3. Compare YOLO, SSD, and Faster R-CNN based on speed, accuracy, and practical applications.
4. What is a bounding box? Why is it important in object detection?
5. Explain the significance of confidence score and Intersection over Union (IoU) in object detection.
6. What are the advantages of using pre-trained deep learning models over traditional image processing techniques?
7. Why are datasets such as MS COCO and Pascal VOC widely used for object detection research?
8. Mention five real-world applications of object detection in computer vision.
9. What challenges are commonly encountered while detecting objects in complex real-world environments?
10. How can object detection systems be further improved using recent advancements in deep learning and computer vision?